

# Yashvardhan SHARMA



## PROFILE

A Dynamic data science professional experienced in on-device ML. Previous research focused on mitigating bias and enhancing robustness in neural networks for computer vision. Seeking roles in AI, Machine Learning, Computer Vision, and NLP.

## EDUCATION

### Indian Institute of Technology Roorkee

B.Tech Production & Industrial Engg. 2022

## CONTACT DETAILS

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yashvardhan-linkedIn

## AWARDS AND ACHIEVEMENTS

Chanakya Fellowship  
Silver Medal at Inter-IIT Tech Meet

## SKILLS

- Python, Kotlin
- Pytorch, Pandas, SKLearn, Numpy
- ONNX-Runtime, Tensorflow-lite, MNN

## REFERENCES

### Prof Vinay Namboodiri

vpn22@bath.ac.uk  
Department of Computer Science  
University of Bath, London.

### Prof Dhish Saxena

dhish.saxena@me.iitr.ac.in  
Mechanical and Industrial  
Department  
Indian Institute of Technology  
Roorkee

## WORK EXPERIENCE

### ML ENGINEER at Nimbleedge AI

June 2022-Present

#### Federated ML Research and Engineering

- ◇ Conducted extensive research on on-device ML, specializing in Federated Learning (FL), real-time-ML, hyper-personalization through fine-tuning, and cohort-wise clustering techniques.
- ◇ Developed in-house FL and hyper-personalization libraries for performance simulations, pioneering novel hybrid personalization techniques which improved user metrics by **4-6 %** on standard datasets.

#### On Device Model Execution

- ◇ Facilitated seamless integration between clients' ML teams and our platform through optimized on-device execution flows.
- ◇ Implemented robust pipelines for on-device model execution, encompassing model conversion, custom operations, and pre/post-processing scripts for diverse applications including text-to-speech (TTS), speech-to-text (ASR) of GenAI and object detection, classification of computer vision.
- ◇ Our deployed real-time on-device model improved performance metrics by **2-4%** in production averaged across millions of users.
- ◇ Developed a large-scale comprehensive mobile testing framework for on-device ML execution to ensure prediction sanity, latency optimization, and memory management. This involved significant input into both the Android SDK and the Python driver orchestrating the tests.
- ◇ Use of this framework helped us identify limits of on-device ML along with several issues across different implementation layers.

## RESEARCH EXPERIENCE

### UNDERGRADUATE THESIS at Department of Computer Science, University of Bath (London)

Jan 2021-May 2021

- ◇ Research focused on fairness in Machine Learning in adversarial framework.
- ◇ Developed a novel masking scheme based on gradient based activations(GBA) to address various biases present in the models.
- ◇ Achieved State-of-the-art results on existing benchmark datasets.
- ◇ [Research paper](#) accepted at **AAAI'22** (an A\* ML conference).

### RESEARCH INTERN at School of Mathematics and Statistics, UNSW, Sydney (Australia)

Jan 2021-May 2021

- ◇ Implemented various evolutionary algorithms including G3PCX, DE, PSO.
- ◇ Developed island versions for population diversity along with use of surrogate NNs for modelling cost to improve computational efficiency for expensive simulations. Achieved state-of-the-art performance on various benchmark problems.
- ◇ [Research paper](#) accepted at **Computational Geosciences'23**.

## PUBLICATIONS

- ◇ Vinod K Kurmi\*, Rishabh Sharma\*, **Yash Vardhan Sharma\***, Vinay P Namboodiri (\*equal contributions), "Gradient Based Activations for Accurate Bias-Free Learning", **Proceedings of the AAAI Conference on Artificial Intelligence, (AAAI), Vancouver BC, Canada, Feb 2022**.
- ◇ Rohitash Chandra, **Yash Vardhan Sharma**, "Surrogate-assisted distributed swarm optimisation for computationally expensive models" in **Computational Geosciences**.

## CAMPUS PROJECTS

### ML REPRODUCIBILITY CHALLENGE

Oct 2020 - Jan 2021

- ◇ Implemented and thus verified the claims of the [ICML 2020 paper](#) on uncertainty quantification in deep neural networks. We provided detailed ablation of hyperparameters and inductive-priors of the algorithm along with proposing an extension for explicit disentanglement of uncertainties.
- ◇ Work was appreciated by the authors, report available [here](#).
- ◇ We applied our proposed extension for the **BOSCH TRAFFIC SIGN RECOGNITION at Inter IIT Tech Meet 2021** clubbed with a Spatial Transformer Networks (STN), hence developing an interpretable uncertainty aware system.
- ◇ Our team bagged **Silver Medal** in this event.